Unit outline_

COMP5046: Natural Language Processing

Semester 1, 2026 [Normal evening] - Camperdown/Darlington, Sydney

On this page

- Overview
- Assessment
- Learning support
- Weekly schedule
- Learning outcomes
- Responding to student feedback
- Additional information

Useful links

- Student Portal →
- Timetables →
- Key dates →
- Canvas →
- Student Admin →
- Health management →

Overview

This unit introduces the computational methods, methodological concepts, and theoretical ideas that underpin modern methods for processing natural language (such as English) using computers. NLP is used in a wide range of applications, including information retrieval and extraction; question answering; machine translation; code generation; dialogue; and classifying and clustering of documents. To achieve these, NLP systems perform a range of tasks using mathematical representations of language. In the context of these applications, the unit will explore common modelling methods, including heuristics, linear models, and neural networks, up to and including large language models (LLMs). The unit covers key ideas relevant to NLP from machine learning, statistics, linguistics, and data science. Students will implement NLP systems and evaluation metrics in labs and assignments. The unit will also investigate the annotation process for creating evaluation data for NLP systems. Students will annotate data as part of completing a real-world NLP task.

Unit details and rules

Academic unit	Computer Science
Credit points	6
Prerequisites ?	None
Corequisites ?	None
Prohibitions ?	COMP4446
Assumed knowledge ?	Knowledge of an OO programming language
Available to study abroad and exchange students	Yes

Teaching staff

Coordinator	Jonathan Kummerfeld, jonathan.kummerfeld@sydney.edu.au

Assessment

ⓘ The census date for this unit availability is **31 March 2026**

Details | Criteria

Type	Description	Weight	Due	Length	Use of AI
Written exam	**Final exam** Written exam	60%	Formal exam period	2 hours	AI prohibited
Outcomes assessed: LO5 LO6 LO7 LO1 LO2 LO3 LO4					
Out-of-class quiz	**Online quizzes** Short online exam-style questions	3%	Multiple weeks	10 minutes	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4					
Practical skill	**Word Vectors** Implementing and evaluating word vectors	10%	Week 02 **Due date**: 06 Mar 2026 at 23:00 **Closing date**: 13 Mar 2026	Code of variable length	AI allowed
Outcomes assessed: LO1 LO2 LO4 LO5					
Practical skill	**Classification Models** Implementing and evaluating classification models	5%	Week 04 **Due date**: 20 Mar 2026 at 23:59 **Closing date**: 27 Mar 2026	Code of variable length	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4 LO5					
Practical skill	**Sequence Models** Implementing and evaluating sequence models	10%	Week 07 **Due date**: 17 Apr 2026 at 23:59 **Closing date**: 24 Apr 2026	Code of variable length	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4 LO5					
Practical skill 👥	**Data Annotation** Performing and evaluating the data annotation process	10%	Week 09 **Due date**: 01 May 2026 at 23:59 **Closing date**: 08 May 2026	Code of variable length	AI allowed
Outcomes assessed: LO6 LO1 LO4 LO5					
Written test 🚧	**Core Concepts Test** This test covers core concepts that students must understand in order to pass the unit. It is a hurdle task with a requirement of 80%. There is an opportunity to try again in week 13.	0%	Week 12	50 minutes	AI prohibited
Outcomes assessed: LO1 LO2 LO3 LO4 LO5 LO6 LO7					
Practical skill	**Lecture tasks** Questions to promote synthesis of concepts	2%	Weekly	n/a	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4					

🚧 = hurdle task ? 👥 = group assignment ?

Assessment summary

Weekly lecture tasks, online quizzes, four assignments, a core concept test (hurdle task), and a final exam.

Use of generative artificial intelligence (AI)

You can use generative AI tools for open assessments. Restrictions on AI use apply to secure, supervised assessments used to confirm if students have met specific learning outcomes.

Refer to the assessment table above to see if AI is allowed, for assessments in this unit and check Canvas for full instructions on assessment tasks and AI use.

If you use AI, you must always acknowledge it. Misusing AI may lead to a breach of the Academic Integrity Policy.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Late submission

In accordance with University policy, these penalties apply when written work is submitted after 11:59pm on the due date:

- Deduction of **5%** of the maximum mark for each calendar day after the due date.
- After ten calendar days late, a mark of zero will be awarded.

This unit has an exception to the standard University policy or supplementary information has been provided by the unit coordinator. This information is displayed below:

This unit limits late submissions, with a maximum of 5 days late over all assignments and a maximum of 2 days late per assignment. There is no penalty within those constraints.

Academic integrity

The University expects students to act ethically and honestly and will treat all allegations of academic integrity breaches seriously.

Our website provides information on academic integrity and the resources available to all students. This includes advice on how to avoid common breaches of academic integrity. Ensure that you have completed the Academic Honesty Education Module (AHEM) which is mandatory for all commencing coursework students

Penalties for serious breaches can significantly impact your studies and your career after graduation. It is important that you speak with your unit coordinator if you need help with completing assessments.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Learning support

Simple extensions

If you encounter a problem submitting your work on time, you may be able to apply for an extension of five calendar days through a simple extension. The application process will be different depending on the type of assessment and extensions cannot be granted for some assessment types like exams.

Special consideration

If exceptional circumstances mean you can't complete an assessment, you need consideration for a longer period of time, or if you have essential commitments which impact your performance in an assessment, you may be eligible for special consideration or special arrangements.

Special consideration applications will not be affected by a simple extension application.

Using AI responsibly

Co-created with students, AI in Education includes lots of helpful examples of how students use generative AI tools to support their learning. It explains how generative AI works, the different tools available and how to use them responsibly and productively.

Support for students

The Support for Students Policy reflects the University's commitment to supporting students in their academic journey and making the University safe for students. It is important that you read and understand this policy so that you are familiar with the range of support services available to you and understand how to engage with them.

The University uses email as its primary source of communication with students who need support under the Support for Students Policy. Make sure you check your University email regularly and respond to any communications received from the University.

Learning resources and detailed information about weekly assessment and learning activities can be accessed via Canvas. It is essential that you visit your unit of study Canvas site to ensure you are up to date with all of your tasks.

If you are having difficulties completing your studies, or are feeling unsure about your progress, we are here to help. You can access the support services offered by the University at any time:

Support and Services (including health and wellbeing services, financial support and learning support)
Course planning and administration
Meet with an Academic Adviser

Weekly schedule

WK	Topic	Learning activity	Learning outcomes
Week 01	Introduction and Word Vectors	Lecture (2 hr)	LO1 LO2 LO4
	Introduction and Word Vectors	Workshop (1 hr)	LO5
Week 02	Foundation of NLP Systems	Lecture (2 hr)	LO1 LO2 LO3 LO4
	Foundation of NLP Systems	Workshop (1 hr)	LO5
Week 03	Models - Non-linear	Lecture (2 hr)	LO3
	Models - Non-linear	Workshop (1 hr)	LO5
Week 04	Inference - Greedy and Search	Lecture (2 hr)	LO1 LO2 LO4
	Inference - Greedy and Search	Workshop (1 hr)	LO5
Week 05	Models - Encoder-decoder	Lecture (2 hr)	LO3 LO4
	Models - Encoder-decoder	Workshop (1 hr)	LO5
Week 06	Models - Transformer	Lecture (2 hr)	LO3
	Models - Transformer	Workshop (1 hr)	LO5
Week 07	Models - Large Language Models	Lecture (2 hr)	LO3
	Models - Large Language Models	Workshop (1 hr)	LO5
Week 08	Data - Annotation and crowdsourcing	Lecture (2 hr)	LO1
	Data - Annotation and crowdsourcing	Workshop (1 hr)	LO5
Week 09	Training - Unsupervised	Lecture (2 hr)	LO1 LO3 LO4
	Training - Unsupervised	Workshop (1 hr)	LO5
Week 10	Models - Agents	Lecture (2 hr)	LO2 LO3
	Models - Agents	Workshop (1 hr)	LO4
Week 11	Training - Reinforcement Learning	Lecture (2 hr)	LO2 LO3
	Training - Reinforcement Learning	Workshop (1 hr)	LO5
Week 13	Advanced Topics	Lecture (2 hr)	LO1
	Advanced Topics	Workshop (1 hr)	LO5

Study commitment

Typically, there is a minimum expectation of 1.5-2 hours of student effort per week per credit point for units of study offered over a full semester. For a 6 credit point unit, this equates to roughly 120-150 hours of student effort in total.

Learning outcomes

Learning outcomes are what students know, understand and are able to do on completion of a unit of study. They are aligned with the University's graduate qualities and are assessed as part of the curriculum.

Outcomes | Graduate qualities | Professional standards

At the completion of this unit, you should be able to:

LO1. apply basic linguistic knowledge to identify properties of text
LO2. understand the internal architecture of language models including the purpose of each component
LO3. implement and train machine learning based systems for solving natural language tasks
LO4. evaluate the performance of natural language processing systems
LO5. mplement and debug a large NLP system in a collaborative manner
LO6. annotate data using appropriate quality control methods
LO7. identify ethical concerns in NLP systems and ways to mitigate those issues

Responding to student feedback

This section outlines changes made to this unit following staff and student reviews.

In response to student feedback, the large project has been removed and some of its parts have been added to earlier assignments instead.

Additional information

Additional costs

Rather than buying a textbook, students will be expected to pay to query certain API services and cloud computing resources.

Disclaimer

Important: the University of Sydney regularly reviews units of study and reserves the right to change the units of study available annually. To stay up to date on available study options, including unit of study details and availability, refer to the relevant handbook.

The information included on this page is correct at 31 Jan 2026.

To help you understand common terms that we use at the University, we offer an online glossary.

We recognise and pay respect to the Elders and communities - past, present, and emerging - of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

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